

Dear Editors,

Please find enclosed our submission entitled

“Boyd’s conductor-11 Mahler measure conjecture: proof of the split-integral identity (C3), with an exact structural analysis of the family S_k ”

by Huimin Zheng (College of Information and Network Engineering, Anhui Science and Technology University, Fengyang, Anhui 233100, P. R. China; zhhm@ahstu.edu.cn), which we submit for consideration in the *Journal of Number Theory*.

This manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors (a single author) are aware of and approve this submission.

Summary and significance. Boyd’s 1998 tables of conjectural identities between Mahler measures of two-variable polynomials and L -values of elliptic curves begin with the smallest possible conductor, $N = 11$. Of the three conductor-11 identities, the first two were proved by Brunault; the third, (C3), concerns a polynomial vanishing on the unit torus and asserts that a signed split integral of $\log |y|$ around the branch cut equals $b_{11} = L'(E_{11}, 0)$. This paper proves (C3), completing the conductor-11 block of Boyd’s table. The main contributions are:

- A closed-cycle construction: Samart’s signed open chain is closed by a small-branch compensating arc into an anti-invariant integral cycle, whose homology class is proved to be twice a generator of the anti-invariant homology; the period ratio is a-priori an integer and is pinned to 2 by certified interval arithmetic (Arb ball arithmetic with rigorous error bounds).
- A direct regulator computation via Brunault’s proved Siegel-unit formula — the symbol $\{x, y\}$ is a pair of modular units on $X_1(11)$, so Bloch’s diamond theorem (and its normalization ambiguities) is entirely bypassed.
- A complete structural analysis of the family S_k : exact determination of the torus intersections, the unique modular-unit case, and the tempered cases, together with a mechanistic counterexample at conductor 53 showing precisely where the closed-cycle/modular-unit mechanism provably fails.

We believe the paper fits the *Journal of Number Theory* well: it resolves a long-standing open identity at the head of Boyd’s tables in the arithmetic of Mahler measures and elliptic-curve L -values, a topic with a strong publication history in JNT.

Transparency. The research was carried out by the author with the assistance of the AI system Kimi (Moonshot AI); a formal Declaration on the use of AI tools appears on the title page of the article, and the author has verified all mathematical content and takes full responsibility for it. All certification scripts are openly available at <https://github.com/huiminZheng-collab/boyd-conductor11> and archived at <https://doi.org/10.5281/zenodo.21820650>. A preprint has been submitted to arXiv (identifier to be announced; available on request).

Suggested referees.

- Matilde N. Lalin (Université de Montréal)
- François Brunault (ENS de Lyon)
- Mathew Rogers
- Wadim Zudilin (Radboud University)

- Detchat Samart
- Marie-José Bertin

The author has no conflicts of interest with the suggested referees.

Thank you for your consideration.

Sincerely,

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